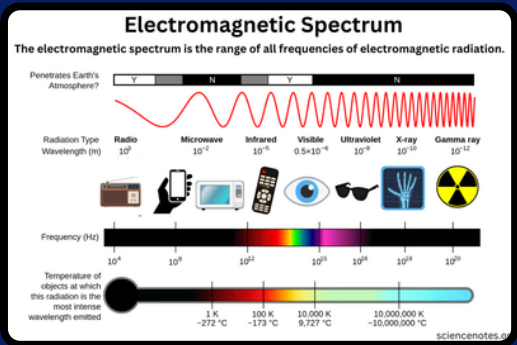


REMOTE CONTROL

INFOGRAPHIC



Frequency and Electromagnetic Spectrum

Frequency is the measure of the amount of cycles that have been undergone in one unit of time. It is often measured in Hz. The electromagnetic spectrum would be the total amount of frequencies of electromagnetic radiation going from minimum 1Hz radio waves to 10^{25} Hz Gamma Rays.

Electromagnetic Spectrum Devices

Radio Waves (Range: 10^1 - 10^9 Hz): Radios, Televisions, Remote-Controlled Toys

Visible Light (Range: 10^{14} - 10^{15} Hz): Cameras, Lightbulbs, Device Screens

Microwaves (Range: 10^9 - 10^{12} Hz): Microwave Ovens, Phones, Laptops, Satellites

Ultraviolet (Range: 10^{15} - 10^{17} Hz): Tanning Beds, UV Lamps

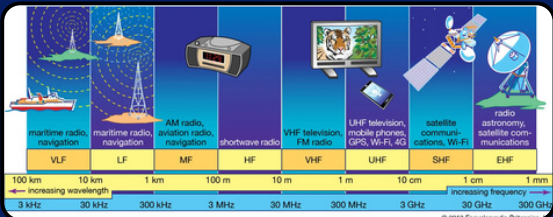
Infrared (Range: 10^{12} - 10^{14} Hz): TV Remotes, Night Vision Cameras

X-Rays (Range: 10^{17} - 10^{19} Hz): Medical X-Ray Imaging, Security Scanners

Gamma Rays (Range: $>10^{19}$ Hz): Radiation Therapy, Astronomy, Sterilization

Radio Frequency Spectrum and Radio Broadcasting

The radio frequency spectrum is a section of the electromagnetic spectrum primarily used for communication and transmitting signals. The frequencies range from 3 Hz to 300 GHz. Radio Broadcasting is the transmission of sound over electromagnetic waves. It is achieved by converting sound to electrical signals through transmitters which are then turned into electromagnetic waves through an antenna in order to be captured and turned back to sound by a receiver.



Radio Spectrum Sections and Devices

The radio spectrum is split up into 12 different sections. These sections or “bands” are: Extremely Low Frequency (ELF, 3-30Hz), Super Low Frequency (SLF, 30-300Hz), Ultra Low Frequency (ULF, 300-3000Hz), Very Low Frequency (VLF, 3-30kHz), Low Frequency (LF, 30-300kHz), Medium Frequency (MF, 300-3000kHz), High Frequency (HF, 3-30MHz), Very High Frequency (VHF, 30-300MHz), Ultra High Frequency (UHF, 300-3000MHz), Super High Frequency (SHF, 3-30GHz), Extremely High Frequency (EHF, 30-300GHz) and Terahertz/Tremendously High Frequency (THF, 300-3000GHz). Device examples for each band include:

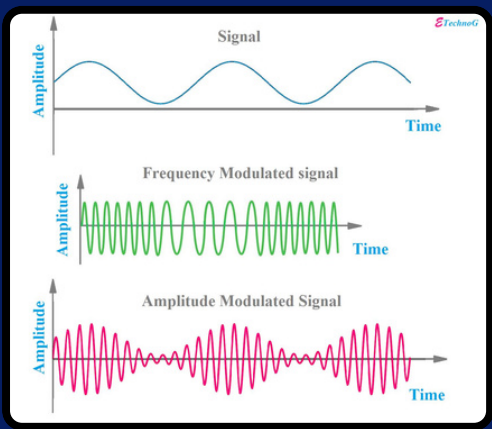
ELF: Submarine Communication and Earthquake Forecasting
SLF: Submarine Communication and Pipeline Maintenance
ULF: Military Communication and Mining
VLF: Submarine Communication and Navigation
LF: Navigation Beacons and Time Signals
MF: AM Radio Broadcasting
HF: Shortwave Radio and Aviation Communication
VHF: FM and Weather Radio Frequencies, Television
UHF: TV's, GPS, Mobile Phones, Wi-Fi and Family Radio
SHF: Radar and Satellite Communication
EHF: Advanced Radar, Experimental Tech and 5G networks
THF: Medical Diagnostics, Security Screening and 6G networks

Transmitters and Receivers

A transmitter is a device that generates and sends a signal from one place to another. A transmitter works by taking a signal, modulating it into a high-frequency wave, amplifying it with an amplifier, adjusts it to the correct frequency and is then converted into electromagnetic waves by the antenna and released into the air. A receiver is a device that accepts the radio waves in the air and decodes them back into usable information. They work by taking in a electromagnetic wave through the antenna, which converts it back to an electrical signal. After which, the signal is amplified and demodulated back into the original signal, which is then ready for output. These functionalities are what make transmitters and receivers critical to the operation of a remote controlled system, since they are how the system is able to communicate remotely.

Channel Mapping

Channel mapping refers to the corresponding output on the receiver that is assigned to a particular stick command on the transmitter. The amount of channels correspond to the amount of outputs that can be executed on the remote control system. The channel mapping is important because it defines the order in which the inputs from the transmitter control the outputs on the receiver. If the channel mapping is wrong, then the joystick inputs will correspond to the wrong outputs (i.e., steering stick controlling throttle), making the remote control system difficult to control.



Modulation Basics

Modulation is the process of tweaking carrier signals increase the distance and clarity that information signals can travel with. The two main forms of modulation include Amplitude Modulation (AM) and Frequency Modulation (FM). Amplitude Modulation changes the carrier wave's strength (amplitude) based on the ups and downs the information signal while keeping its frequency constant. This primarily used to facilitate the travel of information through longer distances, even at lower frequencies. Frequency Modulation changes the frequency of a carrier wave while keeping its amplitude steady. FM is used to deliver higher audio quality and reduce noise when transferring information signals, eliminating distortion, especially compared to AM.

Simple Remote Control System

Function of a remote control center can be condensed into the role of four components: the button, the transmitter, the receiver and the actuator. The button is physical input on the remote; it creates a specific signal based on the corresponding press. The transmitter then takes the input signal and then converts and amplifies it into an electromagnetic wave. The wave is then sent to the receiver, which receives the signal and amplifies and demodulates the signal into its original intended electrical signal. The signal is then sent to the actuator, the physical component that performs an action based on the corresponding signal that was received.



Application

- Drones use Radio Frequency (RF) for wireless communication of audio and video signals. They use a transmitter and receiver to send radio waves for remote communication. Drones use Frequency Hopping to ensure clean and clear communication with the remote. Frequency hopping involves the transmitter and receiver deciding on an pseudo-random sequence of channels to switch between. The transmitter then sends short signals across the channel bandwidth, with the receiver choosing the best signal for demodulation, typically resulting in better performance than non-frequency hopping receivers.
- The difference between radio frequency remote control systems and infrared remote control systems is the frequency range of each system. Radio frequency uses radio waves ranging from 3Hz to 300GHz while infrared systems utilize infrared light with frequencies from 300GHz to 400THz. Radio waves are preferred for applications that do not have line-of-sight and require longer range, greater security or greater data rates (i.e. wireless router). Infrared remote control systems are better suited for applications that have line-of-sight and require a lower budget due to being a cheaper technology (i.e. TV remotes).

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